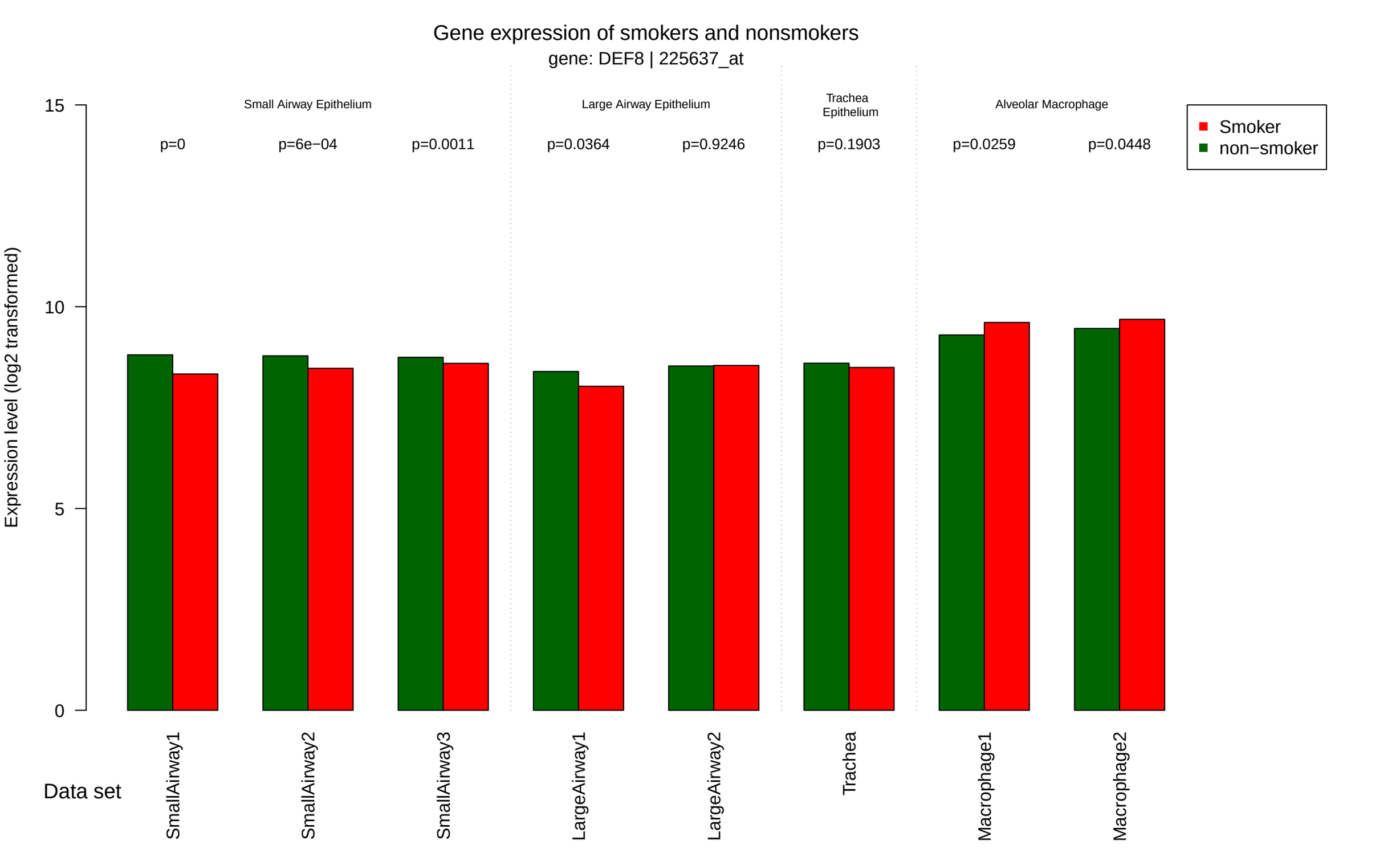
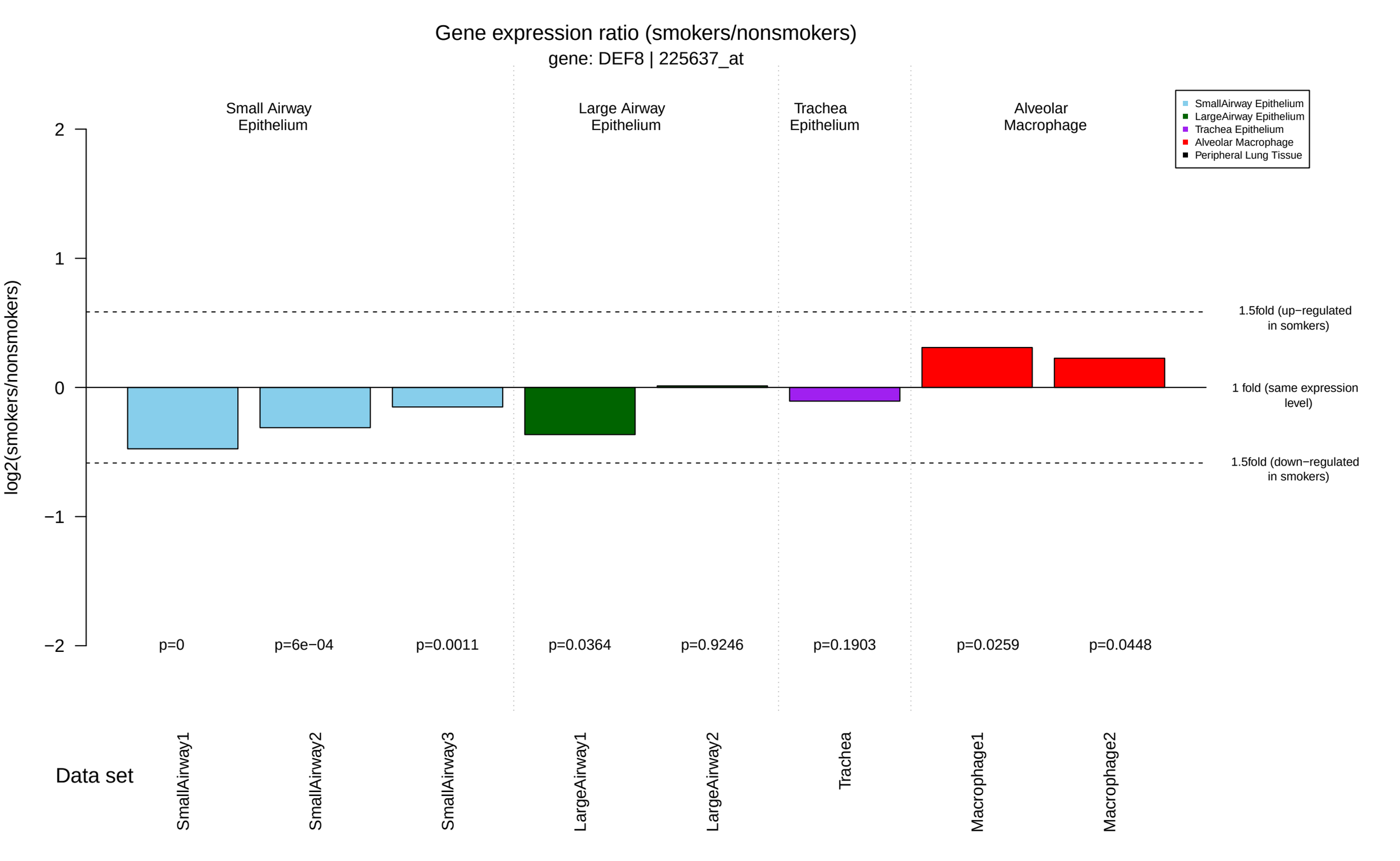
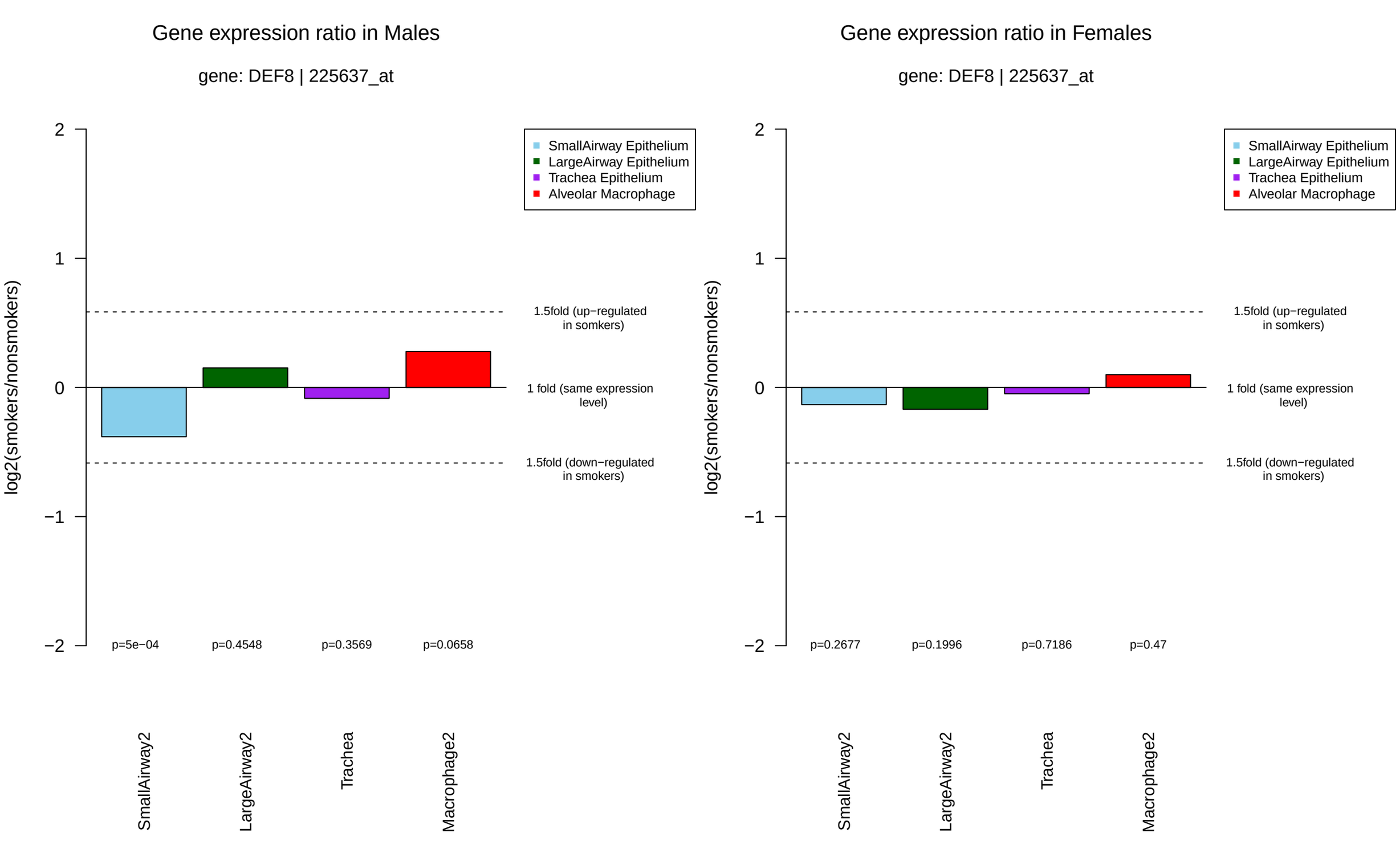
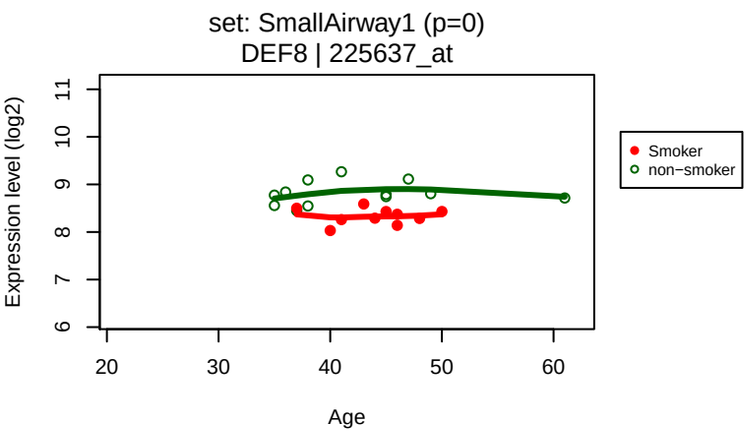
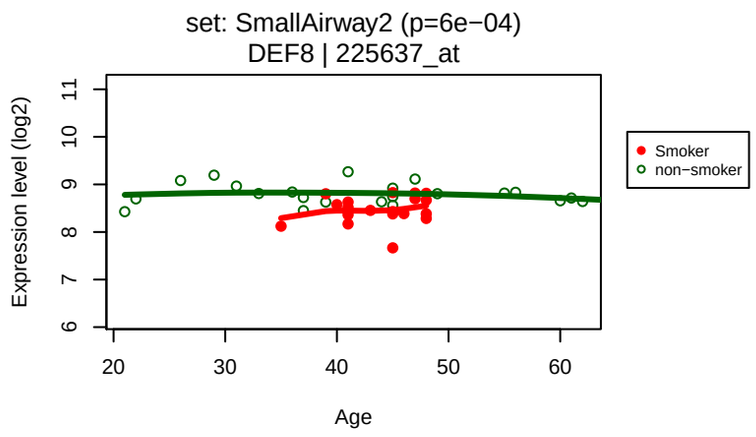
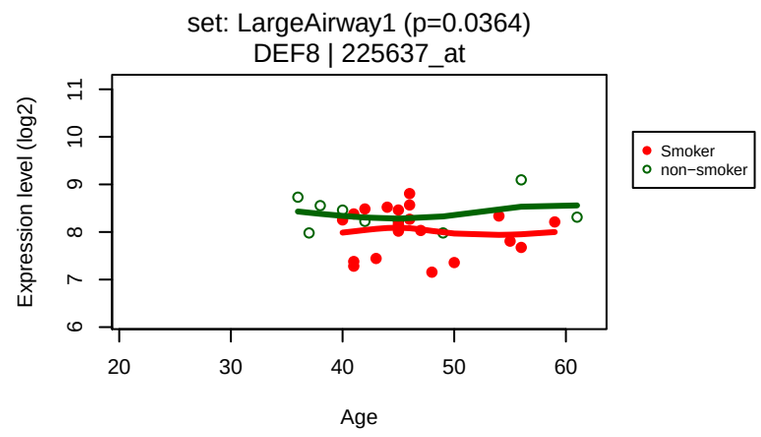

Gene expression of smokers and nonsmokers
gene: DEF8 | 225637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0
p=6e−04
p=0.0011
p=0.0364
p=0.9246
p=0.1903
p=0.0259
p=0.0448
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

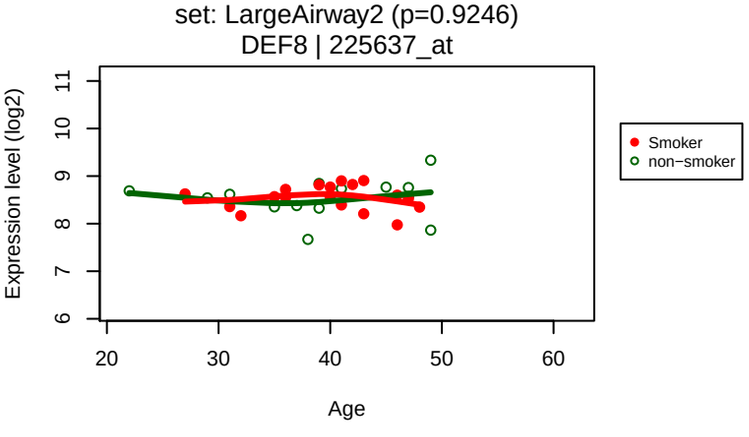

Gene expression ratio (smokers/nonsmokers)
gene: DEF8 | 225637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0
p=6e−04
p=0.0011
p=0.0364
p=0.9246
p=0.1903
p=0.0259
p=0.0448
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

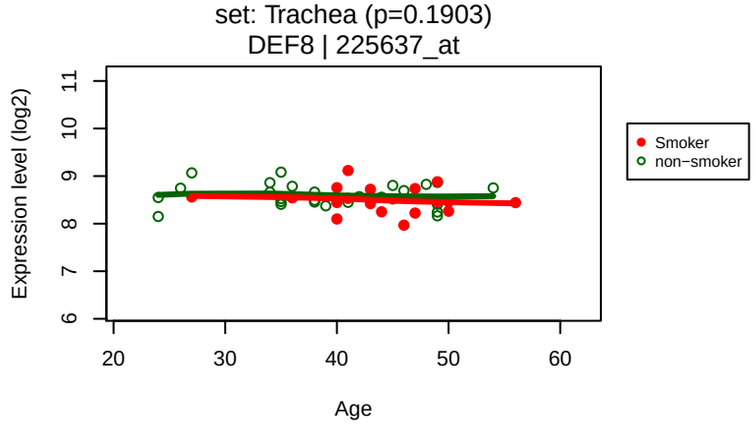

Gene expression ratio in Males
gene: DEF8 | 225637_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=5e−04
p=0.4548
p=0.3569
p=0.0658
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DEF8 | 225637_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2677
p=0.1996
p=0.7186
p=0.47
SmallAirway2
LargeAirway2
Trachea
Macrophage2

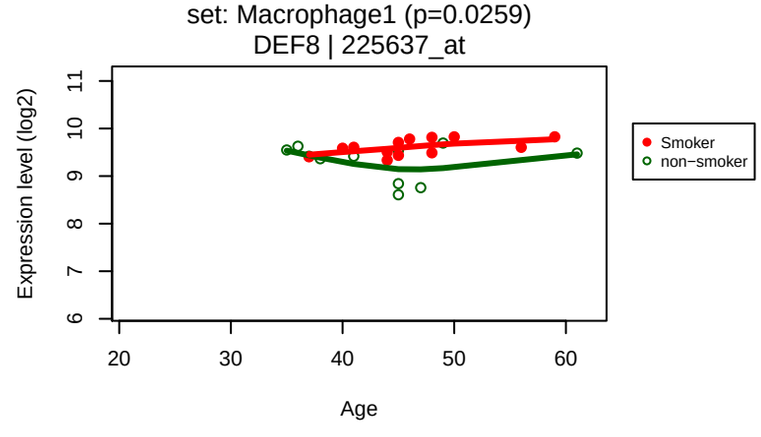

set: SmallAirway1 (p=0)
DEF8 | 225637_at
Expression level (log2)
Age
Smoker
non−smoker

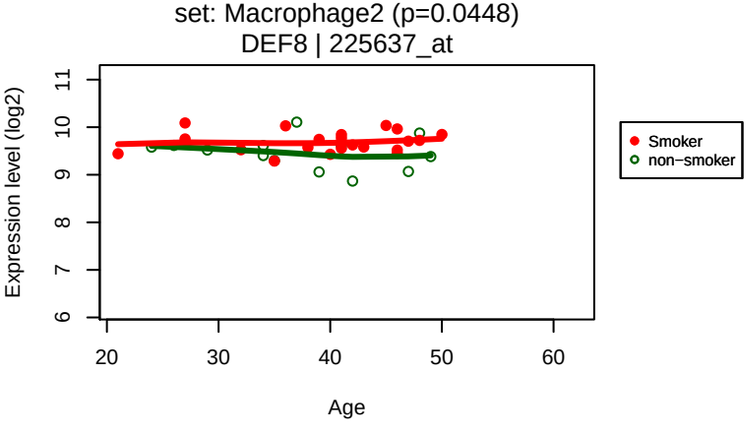

set: SmallAirway2 (p=6e−04)
DEF8 | 225637_at
Expression level (log2)
Age
Smoker
non−smoker

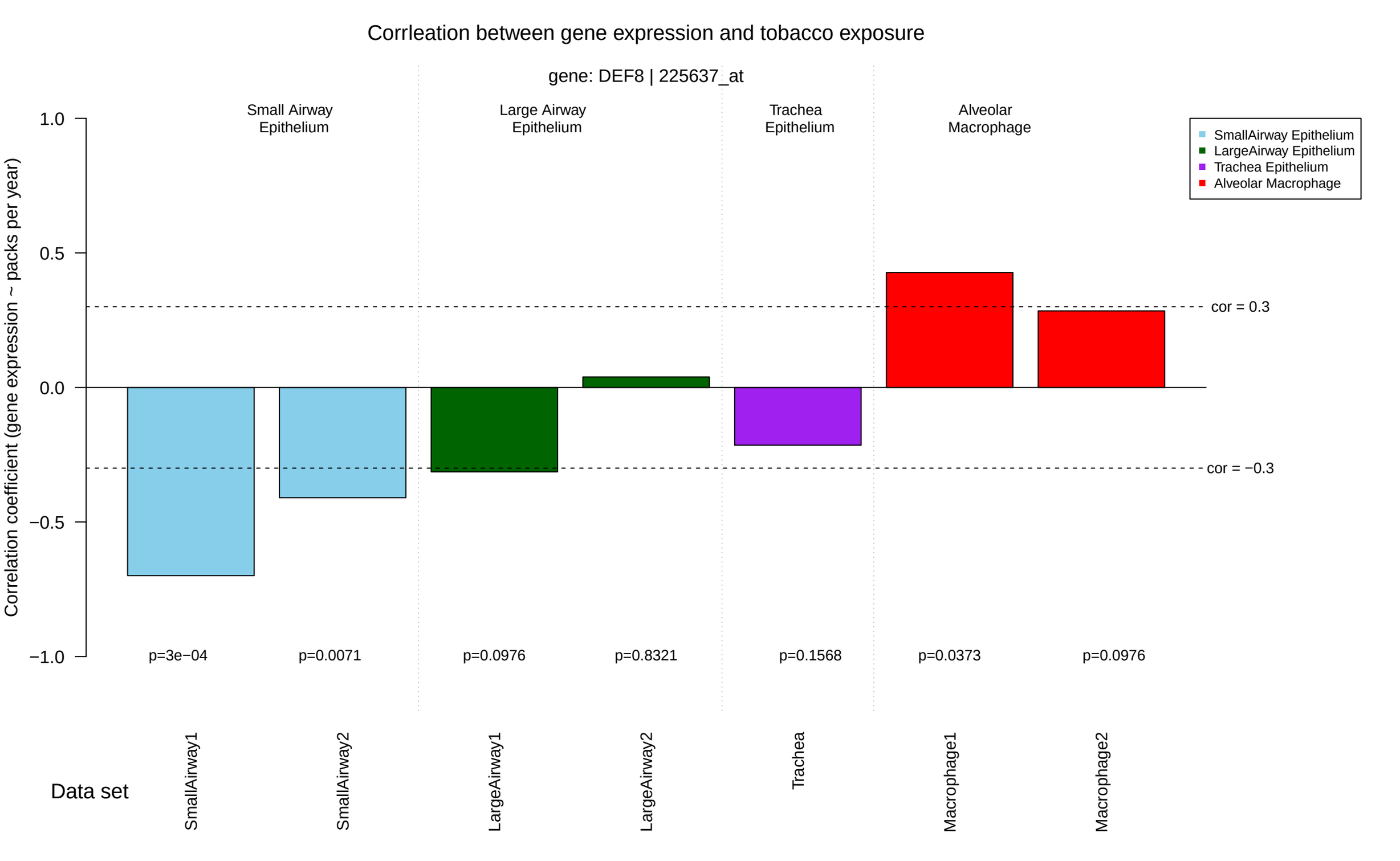

set: LargeAirway1 (p=0.0364)
DEF8 | 225637_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9246)
DEF8 | 225637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1903)
DEF8 | 225637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0259)
DEF8 | 225637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0448)
DEF8 | 225637_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DEF8 | 225637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=3e−04
p=0.0071
p=0.0976
p=0.8321
p=0.1568
p=0.0373
p=0.0976
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2